

Feeling of Control for Virtual Object Manipulation in Handheld AR

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ABSTRACT

One essential feature of handheld augmented reality (AR) is the ability to manipulate virtual objects employing different interaction techniques, such as touch-based, mid-air and tangible interactions. A critical factor for user experience evaluation of AR systems is users' feeling of control. However, little was known in literature about feeling of control for virtual object manipulation in handheld AR. This study explores users' feeling of control among three interaction techniques via self-report and task performance. The findings reveal that users perceived a stronger feeling of control when applying tangible interaction technique to manipulate virtual objects than other techniques in handheld AR.

Keywords: Handheld augmented reality, feeling of control, virtual object manipulation.

Index Terms: H.1.2 [Human-centered computing]: Human computer interaction (HCI) — HCI design and evaluation methods — User studies.

1 INTRODUCTION

Augmented reality (AR) is the presentation of computer-generated objects integrated with actual surroundings. It can be implemented on handheld devices, known as handheld AR, making it accessible to users in their daily lives. Virtual object manipulation is an essential feature in handheld AR [1]. Some interaction techniques have been proposed for virtual object manipulation in handheld AR [1]. Touch-based interaction is a basic interaction type of handheld AR, as users can directly interact with virtual objects by touching screens. Mid-air interaction enables users to control virtual content on the screen through their own hands below the device camera based on hand recognition. Since handheld AR combines virtual and real contents, the correspondence between real and virtual objects allows tangible interaction, which facilitates the control of virtual contents on the screen with real objects in the physical environment. Prior literature compared touch-based and mid-air interactions in user experience to control virtual objects in handheld AR [1]. However, little was known in literature about the difference between tangible interaction and the other two interactions.

Previous research has focused on the user experience, including task performance and psychological aspects, in AR [2]. Among these psychological aspects, the feeling of control is considered an essential factor in determining the success of AR systems [3]. Prior literature measured the feeling of control via other psychological concepts, such as mental workload and sense of agency [4]. Mental workload is negatively related to users' feeling of control, and sense of agency directly reflects users' feeling of control [4]. Besides, scholars applied self-report along with task performance to evaluate users' feeling of control [5]. The current research on users' feeling of control in handheld AR is insufficient, which is still necessary for further investigation.

Therefore, this study aims to compare users' feeling of control for virtual object manipulation across three interaction techniques (touch-based, mid-air and tangible interactions) in handheld AR through self-report and task performance. The main contributions

of this study are listed as follows: (1) generating new insight into the relationship between users' feeling of control and interaction techniques in handheld AR; (2) providing a deep understanding of three interaction techniques in handheld AR of their benefits and drawbacks on users' feeling of control.

2 USER STUDY

Ten participants (7 males and 3 females) were recruited for this study, with an average age of 23 ± 4 years. Before the experiment, they were informed in detail about this study and signed a consent form. The University Ethics Committee has approved this study. As presented in Figure 1(i), the experimental setup consisted of a 6.1-inch smartphone (iPhone 14 pro), and a physical table for simulating stable AR settings with movable and target objects.

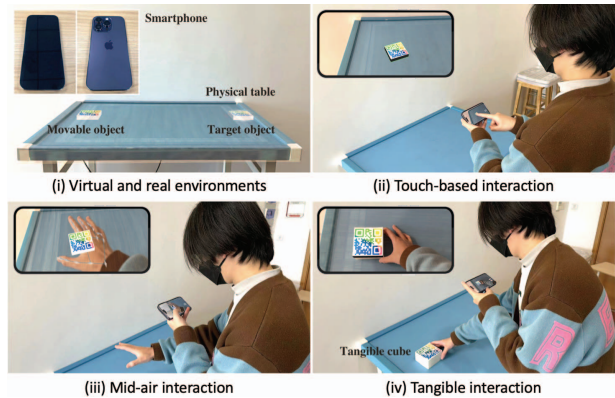


Figure 1: Experimental settings: (i) virtual and real environments, (ii) touch-based interaction, (iii) mid-air interaction and (iv) tangible interaction.

During the experiment, participants were required to complete several tasks with different interaction techniques (touch-based, mid-air and tangible interactions) to move a movable object to a target object in AR settings, as displayed in Figure 1(i).

- Touch-based interaction. As presented in Figure 1(ii), users applied a one-finger gesture to select the movable object on the touchscreen and slide their fingers to move it.
- Mid-air interaction. As displayed in Figure 1(iii), participants employed the right hand corresponding to a virtual hand to touch the movable object, and used one finger of the left hand to click the touchscreen for object selection. Then, participants moved the right hand to manipulate the virtual object.
- Tangible interaction. As shown in Figure 1(iv), participants adopted the right hand to grasp the tangible cube in the actual environment, which was synchronous spatial with the movable object in the virtual world, and they moved the tangible object to synchronize the position of the virtual object.

Before the experiment, participants had a chance to adapt three interaction techniques for virtual object manipulation in handheld AR. Then, they were asked to complete six experimental tasks for each interaction. In each task, a movable object and a target position were placed on a virtual table synchronized with the physical table in actual surroundings. The distance between the moveable object and the target object was the same for each manipulation task in order to maintain the same experimental settings. Participants were

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asked to move the moveable object to the target position, with a color cue given when the moveable object was less than 5 mm from the target. After all tasks with each interaction, participants were required to fill out a questionnaire.

This study evaluated users' feeling of control based on two measures: self-report and task performance. The Wilcoxon signed-rank test was employed to compute whether there were significant differences in these two measures across interaction techniques.

- Self-report. A modified questionnaire was applied with a six-item NASA Task Load Index (TLX) and a four-item Agency Questionnaire for subjective analysis.
- Task performance. The manipulation time and error rate of each task were collected for efficiency and precision analysis.

3 RESULTS

As displayed in Figure 2, significant differences in workload scores were observed between touch-based and mid-air interactions ($\chi^2=5.14$, $p=0.02$), as well as between mid-air and tangible interactions ($\chi^2=6.24$, $p=0.01$). Mid-air interaction ($M=4.50$) was rated with lower workload scores than touch-based ($M=5.79$) and tangible ($M=6.25$) interactions. Besides, significant differences in agency scores were seen between mid-air and tangible interactions ($\chi^2=4.21$, $p=0.04$). Users rated tangible interaction ($M=6.53$) with higher agency scores compared with touch-based ($M=6.19$) and mid-air ($M=5.59$) interactions.

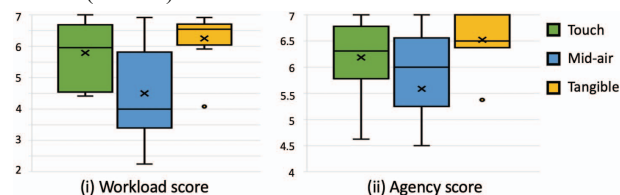


Figure 2: Subjective data: (i) workload score and (ii) agency score.

As presented in Figure 3, significant differences in manipulation time were observed between touch-based and mid-air interactions ($\chi^2=15.27$, $p<0.01$), as well as mid-air and tangible interactions ($\chi^2=15.78$, $p<0.01$). Participants spent more time when employing mid-air interaction ($M=23.01$) than touch-based ($M=12.84$) and tangible ($M=11.76$) interactions. In addition, significant differences in error rate were merely seen between touch-based and mid-air interactions ($\chi^2=10.57$, $p<0.01$). Participants performed lower task precision with mid-air interaction ($M=3.338$) than with touch-based ($M=0.835$) and tangible ($M=2.115$) interactions.

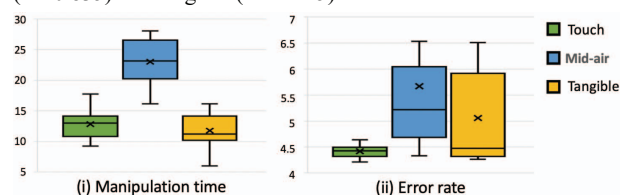


Figure 3: Performance data: (i) manipulation time and (ii) error rate.

4 DISCUSSION

Based on subjective analysis, it is evident that mid-air interaction obtained lower workload scores (increased mental workload) and fewer agency scores (decreased sense of agency) than the other two interactions in handheld AR. Prior literature indicated that mental workload is negatively associated with users' feeling of control, and sense of agency can directly reflect users' feeling of control [4]. Thus, mid-air interaction cannot provide users with a preferable controlling experience for virtual object manipulation in handheld AR. In addition, touch-based and tangible interactions presented the same level of control for users to manipulate virtual objects in

handheld AR, but participants subjectively felt more control when applying tangible interaction than touch-based interaction. Hence, the comparison of the three interaction techniques showed that tangible interaction is preferable to the other two by offering more feeling of control in handheld AR.

In this research, there were no significant differences in task performance between touch-based and tangible interactions, but they both performed significantly better than mid-air interaction in handheld AR. Users spent less time when using tangible interaction on average, which was the most efficient among these interactions. Previous research showed that participants felt a higher feeling of control when they performed efficiently [5, 6]; hence, participants felt a higher feeling of control for virtual object manipulation with less manipulation time when using tangible interaction in handheld AR. Besides, touch-based interaction was observed to be more precise than the other two techniques in handheld AR. The possible explanation is that touch-based interaction only provided users with one-finger gestures in handheld AR, but the other two techniques enabled users to control virtual objects with their own hands, which may make virtual object manipulation with more complicated hand movements. Since literature has seldom explored the link between the feeling of control and task precision in simulated environments, we may undertake further research on this topic.

Generally, according to self-report and performance measures, tangible interaction provided a preferable controlling experience in handheld AR. We recommended tangible interaction for users to manipulate a virtual object in the real-virtual environment, which entirely takes advantage of AR features.

5 CONCLUSION

This study aims to explore users' feeling of control across different interaction techniques (touch-based, mid-air and tangible interactions) for virtual object manipulation in handheld AR. Based on self-report and task performance, the findings of this research reveal that users felt a stronger feeling of control when applying tangible interaction to move virtual objects than other interactions in handheld AR. This study contributes to the current literature by identifying how interaction techniques impact users' feeling of control for virtual object manipulation in handheld AR, as well as analyzing interaction techniques in detail on their advantages and disadvantages when considering feeling of control as an important index of user experience for handheld AR.

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